

Report
Of
FULL STACK DEVELOPMENT

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: 2410031686
Name of Student: MD ASRAR
Batch: 2024–2028

2026–27

CANDIDATE'S DECLARATION

I, **MD ASRAR**, do hereby declare that the internship report titled “Design and Development of a SaaS-Based School Management System Using MERN Stack” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: MD ASRAR

Roll No.: 2410031686

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature: _____

Student Name: MD ASRAR

Roll No.: 2410031686

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INTERNSHIP COMPLETION CERTIFICATE



EARTHIX SOFTWARE SOLUTIONS PVT. LTD.

www.earthix.in | support@earthix.in | 01 9217232660

Full Stack Development Internship

Earthix Software Solutions Pvt. Ltd.

THIS CERTIFICATE IS PROUDLY PRESENTED TO

MD ASRAR

For Successfully Completing the
Full Stack Developer Internship

This certificate is awarded in recognition of the successful completion of the **Full Stack Developer Internship** at **Earthix Software Solutions Pvt. Ltd.**

The internship was successfully completed from **01 June 2026** to **31 July 2026** for a duration of **Two Months (8 Weeks)**.

During the internship period, the intern demonstrated excellent technical skills, dedication, problem-solving ability, and a strong willingness to learn while making valuable contributions to the full stack development projects including frontend, backend, API development and database management.

Employee ID : EX321151

Issued Date : 01/08/2026

Certificate ID : ESSPT321151

Earthix Software Solutions Pvt. Ltd.
Md Sanaullah
Director

Md Sanaullah
Earthix Software Solutions Pvt. Ltd.

Earthix Software Solutions Pvt. Ltd.
Ph : 9217232660

Authorized by
www.earthix.in
Earthix Software Solutions Pvt. Ltd.

4. PROJECT DESCRIPTION

4.1 Introduction

The internship project focuses on the design and development of a SaaS-based School Management System using the MERN stack. The proposed platform is intended to centralize common academic and administrative activities in one web application. A modular approach supports different school functions through role-based access and reusable services. The project emphasizes scalability, maintainability, and a better digital workflow.

4.2 Organization Profile

Earthix Software Solutions Pvt. Ltd. is the organization where the internship was undertaken. The internship role was Full Stack Developer Intern. The work exposure covered frontend development, backend services, databases, APIs, Git/GitHub, and debugging. The internship provided practical exposure to software development workflow and full-stack application development.

4.3 Problem Statement

Schools may rely on disconnected processes for student records, attendance, fees, communication, and academic information. Manual or fragmented workflows can increase repetitive work and make information harder to access. A centralized SaaS platform can provide a single source of operational data and simplify day-to-day management. The project therefore addresses the need for an integrated, web-based school management solution.

4.4 Project Objectives

- Develop a centralized school management web application.
- Implement a responsive and user-friendly frontend using React.js.
- Build RESTful backend services using Node.js and Express.js.
- Use MongoDB for structured and scalable data management.
- Apply role-based access and modular design for different school operations.
- Use Git/GitHub and development tools to support an organized workflow.

4.5 Scope of the Project

Module	Scope
Student Management	Student profiles, records and related information
Attendance	Attendance recording and tracking workflows
Academic Management	Classes, subjects and academic information
Fees & Records	Fee-related records and administrative data
Communication	Centralized information sharing and notifications
Administration	Role-based access and operational management

4.6 Technologies and Tools Used

Technology / Tool	Purpose
HTML & CSS	Structure and styling of web interfaces
JavaScript	Application logic and client-side functionality
React.js	Frontend user interface development
Node.js	Server-side JavaScript runtime
Express.js	Backend API and routing layer
MongoDB	Database for application data
REST APIs	Communication between frontend and backend
Git / GitHub	Version control and source management
VS Code	Development environment
Postman	API testing and verification

4.7 System Architecture

The system follows a layered full-stack architecture. Users interact with the React.js frontend, which communicates with backend REST APIs implemented with Node.js and Express.js. The backend manages application logic and interacts with MongoDB for persistent data storage.

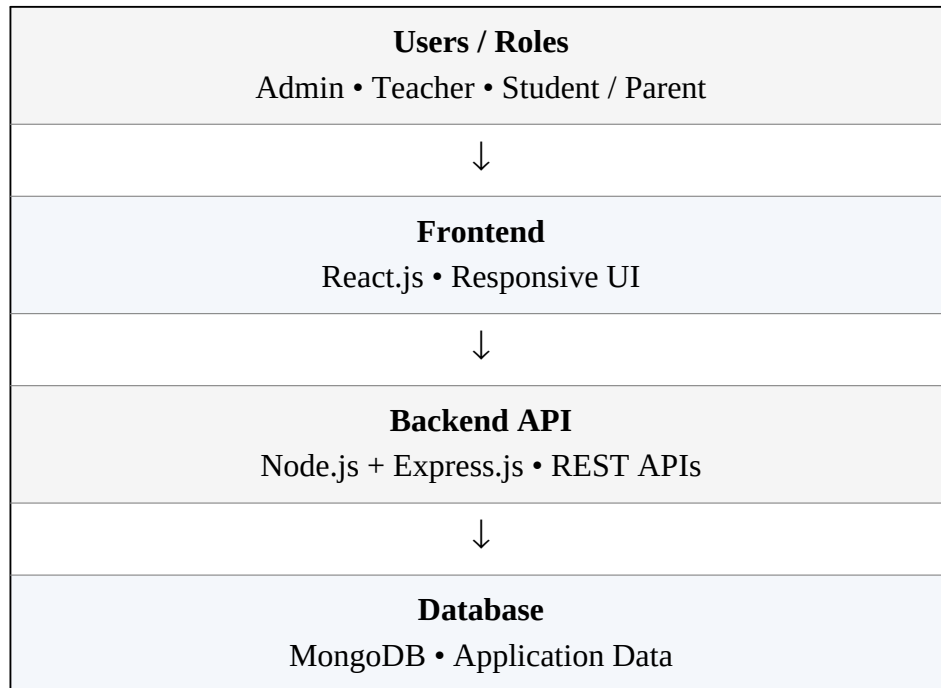


Figure 4.1: High-level system architecture

4.8 Methodology

The development process follows a structured sequence from requirement analysis to testing and version control. The stages are arranged to support incremental development and easier identification of issues.

1	Requirement Analysis
2	UI & Module Planning
3	Frontend Development
4	Backend & API Development
5	Database Integration
6	Testing & Debugging
7	Git/GitHub Version Control

Figure 4.2: Development methodology stages

4.9 Expected Outcomes

- A centralized web-based platform for managing core school operations.
- Improved accessibility of school information through a unified interface.
- Reusable full-stack architecture that can be extended with additional modules.
- Better understanding of production-oriented frontend, backend, database, API, and version-control workflows.
- A foundation that can evolve into a broader SaaS management platform.

4.10 Certificates of Completion and Other Communication Mail Proof

The Internship Completion Certificate issued by Earthix Software Solutions Pvt. Ltd. is attached in this report as required by the prescribed format. The certificate records successful completion of the Full Stack Developer Internship from 01 June 2026 to 31 July 2026 for a duration of two months (8 weeks), with Employee ID EX321151 and Certificate ID ESSPT321151.

The internship offer letter is included as additional supporting internship documentation. The original document is attached on the following pages as communication/proof of the internship engagement.

Additional Internship Communication Proof — Internship Offer Letter



INTERNSHIP OFFER LETTER

Candidate Name:	Md Asrar	Employee ID:	EX321151
Internship Role:	Full Stack Developer Intern	Joining Date:	01/06/2026
Duration:	2 Months	Work Mode:	Work From Home
Stipend:	Unpaid	Contact Number:	+91 8340452870

Dear **Md Asrar**,

We are pleased to offer you the position of **Full Stack Developer Intern** at **Earthix Software Solutions PVT. LTD.**

This internship is designed to provide practical exposure, structured learning, mentorship, and hands-on project experience in Full Stack Web Development within a professional work environment.

During your internship, you will collaborate with the Development, Design, and QA teams, work on assigned full stack deliverables, and gain valuable industry experience in both frontend and backend technologies.

Internship Benefits

- Hands-on practical learning in full stack development (MERN / MEAN Stack)
- Real-world project experience on live website and web applications
- Internship completion certificate
- Letter of recommendation based on performance
- Mentorship from senior full stack developers
- Future employment consideration based on performance and openings



Roles & Responsibilities

- **Frontend Development:** Develop and maintain responsive, user-friendly web interfaces using HTML, CSS, JavaScript, and modern frameworks like React.js / Next.js.
- **Backend Development:** Build and manage server-side logic, APIs, and database operations using Node.js, Express.js, and related backend technologies.
- **Database Management:** Design, implement, and maintain databases (MongoDB / MySQL / PostgreSQL), ensuring data integrity, security, and optimized performance.
- **API Development & Integration:** Create and integrate RESTful APIs, handle third-party API integrations, and ensure seamless communication between frontend and backend.
- **Full Stack Project Execution:** Work on end-to-end development of live projects, from requirement analysis and UI implementation to deployment and maintenance.
- **Version Control & Collaboration:** Use Git/GitHub for version control, collaborate with cross-functional teams, participate in code reviews, and follow agile development practices.
- **Bug Fixing & Optimization:** Debug, test, and optimize applications for maximum speed, scalability, and security across different browsers and devices.
- **Professional Conduct:** Adhere to company policies, maintain confidentiality of project data and source code, meet deadlines, and participate actively in team meetings and reviews.



Terms & Conditions

1. Internship duration is from **1st June 2026 to 31st July 2026** and will continue for 2 Months.
2. This is a Work From Home internship. Intern must have a stable internet connection and laptop/desktop for development activities.
3. All work products, source code, designs, and documentation created during the internship are the intellectual property of Earthix Software Solutions Pvt. Ltd.
4. Confidentiality of all company data, client information, systems, and intellectual property must be strictly maintained even after internship completion.
5. Intern must maintain minimum 90% attendance and submit weekly progress reports to the reporting manager.
6. Unsatisfactory performance, breach of confidentiality, or misconduct may result in immediate termination of internship without certificate.
7. Completion certificate and letter of recommendation are subject to satisfactory performance and completion of assigned tasks.
8. This internship does not constitute an offer of permanent employment, though exceptional performers may be considered for full-time roles.

Acceptance & Confirmation

Candidate Signature: _____

Candidate Name: Md Asrar

Acceptance Date: _____

28-05-2026

Earthix Software Solutions Pvt. Ltd.

Authorized Signatory: Earthix Software Solutions Pvt. Ltd.

Ph : 9217232660
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For more queries, visit our website: www.earthix.in

✉ support@earthix.in

5. BIBLIOGRAPHY / REFERENCES

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